

Disentangling Density from Cycles in Multi-Agent Communication Topologies

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Abstract. At least seven recent multi-agent systems enforce DAG communication topologies, claiming that acyclic structure improves performance. We prove that these claims rest on a methodological confound: for connected graphs with n vertices and m edges, the cycle rank $\beta_1 = m - n + 1$, so every comparison that varies topology while varying edge count conflates cycle structure with density. To escape this identity, we introduce directed simple cycle count $\kappa(D)$ as an experimental variable that decorrelates from density—eight directed graph families at constant $(n, m) = (8, 16)$ span κ from 0 to 47. Across 240 controlled island model GA runs on OneMax, we find a significant main effect of topology on diversity (ANOVA $p < 10^{-7}$, $\eta^2 = 0.17$) with Pearson $r = -0.68$ between κ and diversity at generation 30, with the effect proving transient on OneMax. However, a supplementary NK landscape pilot shows 14 $\times$ amplification at $K = 6$ ($\eta^2 = 0.69$ for diversity), confirming that topology effects scale with landscape ruggedness. The existence of these effects at constant density demonstrates that the DAG hegemony rests on confounded evidence, not controlled experiment.

1 Introduction

In island model genetic algorithms, the *migration topology*—a graph whose vertices are subpopulations and whose edges determine migration routes—shapes diversity maintenance and convergence speed [24, 2]. Ring topologies maintain diversity longer than complete graphs; sparse topologies slow premature convergence [1, 23]. The open question is not *whether* topology matters but *which topological property* is operative. Yang et al. [27] identify topology as the single most underexplored dimension of MAS design, proposing nine research directions; a recent survey formalises agent coordination as graph computation [5]. Neither, however, identifies the algebraic confound we expose below.

Recent LLM-based multi-agent systems (MAS) have intensified this question. Seven major systems—GoAgent [8], AgentConductor [25], Graph-GRPO [7], OFA-MAS [19], G-Designer [28], ARG-Designer [18], and Dochkina et al. [10]—enforce or recommend DAG communication topologies. Against this consensus, BIGMAS [12] permits cycles, Agent Q-Mix [14] learns communication actions that create 2-cycles, and Puppeteer [9] discovers cyclic structures under optimization pressure. The field is split, with no shared framework to adjudicate.

The reason is algebraic. For any connected graph G with n vertices and m edges, the cycle rank $\beta_1(G) = m - n + 1$ [13]: cycle count and edge count differ by a constant. Since Pearson correlation is affine-invariant, $\rho(\beta_1, Y) = \rho(m, Y)$ for any outcome Y . Every comparison of “cyclic” versus “acyclic” topologies therefore simultaneously varies density. The DAG hegemony and pro-cycle findings are both potentially density effects in disguise.

We resolve this via directed graph theory. The simple directed cycle count $\kappa(D)$ is not determined by (n, m) : two digraphs with $n = 8$, $m = 16$ can have 0 or 47 simple directed cycles. Across 240 runs (8 topology families, 30 seeds) at constant density, we find a significant main effect on diversity ($\eta^2 = 0.17$, $p < 10^{-7}$) with $r = -0.68$ between κ and diversity at generation 30.

The contributions are: **(1)** a theorem proving the density–cycle confound for connected undirected graphs; **(2)** directed simple cycle count $\kappa(D)$ as a density-independent experimental variable; **(3)** the first controlled experiment showing that directed cycle structure (κ) has an independent effect on island model GA dynamics; and **(4)** NK landscape results showing 14× amplification on rugged landscapes. While our categorical framework (the confound theorem) motivates the investigation of directed cycle count as a topological invariant, the core contributions of this paper are empirical: controlled experiments demonstrating that topology affects evolutionary dynamics at constant density, that this effect scales with landscape ruggedness, and that diversity maintenance is the primary mediating channel. We view the theoretical explanation of *why* cycles modulate diversity as an open problem for future work.

2 Background

2.1 Island Model Evolutionary Algorithms

In island model EAs, a population is partitioned into semi-isolated subpopulations that evolve independently and exchange individuals via migration [24]. The *migration topology* $G = (V, E)$ determines which islands may exchange migrants. The *density* of G is $|E|/|V|$, the average number of connections per island.¹ Sparse topologies slow the spread of fit individuals, maintaining diversity; dense topologies accelerate convergence but risk premature diversity loss [1, 23].

2.2 Cycle Rank and the First Betti Number

For a graph G with n vertices, m edges, and c connected components, the first Betti number is $\beta_1(G) = m - n + c$. For connected graphs ($c = 1$), $\beta_1 = m - n + 1$. Geometrically, β_1 counts independent cycles: a spanning tree uses $n - 1$ edges, and each additional edge closes exactly one independent cycle [13]. Prior work reports correlations between β_1 and fitness convergence, but as Section 3 shows, these are algebraically indistinguishable from density effects. Recent work on

¹ Some authors use $|E|/\binom{|V|}{2}$; the choice does not affect the confound since both are affine in $|E|$.

Table 1. Systems enforcing the DAG constraint. All vary β_1 and density simultaneously.

System	Claim	Confound
GoAgent [8]	DAG enables tractable CIB	DAG vs. dense baseline varies β_1 and m
Graph-GRPO [7]	Learned DAGs beat complete graphs	Complete = max density and max β_1
OFA-MAS [19]	Sparse communication improves efficiency	Pruning reduces m and β_1 in lockstep
G-Designer [28]	Moderate connectivity best	No density-controlled baseline
AgentConductor [25]	DAG topology evolution for code generation	Evolves DAGs only; cyclic alternatives never tested
ARG-Designer [18]	Autoregressive DAG generation	DAG constraint baked into generative model
Dochkina et al. [10]	44% quality spread ($d=1.86$); capability moderates	Sparsity and acyclicity confounded in all 25K runs

cellular sheaves [11] shows that β_1 can be understood as the dimension of sheaf cohomology $H^1(G; \mathcal{F})$, providing a natural extension to heterogeneous agent capabilities.

2.3 Directed Graphs and Simple Cycle Count

A digraph $D = (V, A)$ has arcs (ordered pairs) rather than edges. The *simple directed cycle count* $\kappa(D)$ is the number of elementary directed cycles in D . Unlike β_1 , which for connected graphs is determined by n and m , $\kappa(D)$ depends on the specific arc arrangement. Enumeration uses Johnson’s algorithm [15].

3 The Density–Cycle Confound

Theorem 1 (Cycle rank–density identity). *Let $G = (V, E)$ be a connected graph with $|V| = n$ and $|E| = m$. Then $\beta_1(G) = m - n + 1$.*

Proof. Any spanning tree of G uses $n - 1$ edges; each remaining edge closes exactly one independent cycle. The result follows from the Euler characteristic $\chi = n - m = 1 - \beta_1$ for connected 1-dimensional simplicial complexes.

Corollary 1. *For connected graphs with fixed n , β_1 and m differ by the constant $n - 1$. Pearson correlation is invariant under affine transformation, so $\rho(\beta_1, Y) = \rho(m, Y)$ for any outcome variable Y .*

The methodological consequence: *any experiment comparing connected topologies on a fixed node count cannot distinguish “cycles affect performance” from “edge count affects performance.”* Table 1 surveys seven systems exhibiting this confound.

3.1 The Directed Escape

Theorem 1 applies to homological cycle rank. For directed graphs, the simple cycle count $\kappa(D)$ is combinatorial, not homological: it depends on arc arrangement, not cycle space dimension. We constructed eight families at constant $(n, m) = (8, 16)$ with κ ranging from 0 to 47. Among strongly connected members alone, κ spans 10 to 47—a $4.7\times$ variation at strictly constant density and identical directed rank 9.

Table 2. Eight directed graph families at constant density ($n = 8$, $m = 16$). κ = directed simple cycle count. SC = strongly connected.

Family	Description	κ	SC
dag-layer	3-layer forward hierarchy	0	No
dag-wide	Fan-out; single coordinator broadcasts	0	No
lowcyc-1	dag-layer + one feedback arc	3	No
bidir-ring	Forward and backward directed 8-ring	10	Yes
two-cliques	Two directed 4-cycles with cross-chords	14	No
mesh-cyclic	2×4 grid with bidirectional verticals	20	Yes
dense-triangles	Overlapping directed triangles + skip-2	29	Yes
ring-skip2	Ring $i \rightarrow i+1$ plus $i \rightarrow i+2 \pmod{8}$	47	Yes

4 Experimental Design

All experiments below use the directed simple cycle count $\kappa(D)$, not the homological β_1 , as the independent variable.

4.1 Graph Families

Table 2 describes the eight directed graph families, all with $n = 8$ vertices and $m = 16$ arcs (density $m/n = 2.0$ held constant). Cycle counts were verified by exhaustive enumeration via Johnson’s algorithm [15].

4.2 Island Model GA

Each vertex hosts one subpopulation of 50 individuals (400 total). The fitness function is OneMax on $L = 100$ -bit strings: $f(\mathbf{x}) = L^{-1} \sum_{k=1}^L x_k \in [0, 1]$. OneMax is unimodal, chosen so that any topology effect is structural, not landscape-specific.

Selection is tournament ($k = 3$); crossover is uniform ($p = 0.5/\text{locus}$); mutation is bit-flip at rate $p_m = 1/L = 0.01$ per bit. Every 10 generations, the 5 fittest individuals on each island migrate along outgoing arcs, replacing the 5 weakest (synchronous migration). All families share density $16/8 = 2.0$. Runs last 500 generations.

4.3 Metrics and Statistical Analysis

Diversity. Mean pairwise normalised Hamming distance across all 400 individuals:

$$D(t) = \frac{1}{\binom{N}{2}} \sum_{i < j} \frac{d_H(\mathbf{x}_i, \mathbf{x}_j)}{L},$$

estimated from 20 randomly sampled pairs when $N > 20$.

Analysis. Pearson r between κ and each metric at generations 30, 60, and 100. One-way ANOVA ($\alpha = 0.05$) with topology as factor; effect size η^2 ; post-hoc Mann–Whitney U with Bonferroni correction. Because all families share $m = 16$, $r(\text{density}, Y) = 0$ identically. Each topology runs with 30 seeds (240 total). Implementation: Haskell; cycle counts pre-computed via Johnson’s algorithm in Python.

4.4 NK Landscape Pilot

To test scaling with landscape difficulty, we ran a pilot on NK landscapes [16] ($N = 100$, adjacent neighbours, $K \in \{0, 4, 6\}$, fixed seed). Three topologies spanning κ : dag-layer (0), bidir-ring (10), ring-skip2 (47); 10 seeds per condition (90 runs).

5 Results

5.1 Topology Significantly Affects Diversity and Fitness

One-way ANOVA yields $F(7, 232) = 6.90$, $p = 1.85 \times 10^{-7}$ at generation 30; Kruskal–Wallis corroborates ($H = 38.53$, $p = 2.40 \times 10^{-6}$). Effect sizes: $\eta^2 = 0.17$ (diversity), $\eta^2 = 0.24$ (fitness)—medium-to-large. Density is constant by construction, so $r(\text{density}, Y) = 0$ identically.

5.2 Cycle Count Negatively Predicts Diversity

Figure 3 shows $\kappa(D)$ against mean diversity at generation 30. Pearson $r = -0.681$ ($r^2 = 0.46$, $p = 0.063$ on 6 df); the marginal p reflects only 8 topology-level points, while the 240-run ANOVA confirms the effect. Post-hoc Bonferroni contrasts: dag-wide ($\kappa = 0$) differs from all $\kappa \geq 10$ topologies ($p < 0.001$).

5.3 The Effect Is Transient

The temporal profile (Figures 1–4): at generation 30, $r = -0.681$, $\eta^2 = 0.17$; at generation 50, $r = 0.838$ for fitness ($p = 0.009$; higher κ slows convergence, yielding lower fitness at generation 50); by generation 100, $|r| < 0.10$, $\eta^2 = 0.002$. OneMax is unimodal, so all topologies eventually converge; κ affects the *rate* of diversity loss, not the outcome. DAG families converge fastest; ring-skip2 ($\kappa = 47$) sustains diversity longest. The bidirectional ring does not fall monotonically, suggesting cycle *length distribution* may contribute independently [22].

5.4 Topology Effect Scales with Landscape Ruggedness

Figure 5 shows η^2 for diversity increasing monotonically with K : from 0.049 ($K = 0$) to 0.445 ($K = 4$) and 0.686 ($K = 6$, both $p < 10^{-3}$) at generation 200. Fitness follows: $\eta^2 = 0.001, 0.337, 0.349$. On the rugged $K = 6$ landscape, nearly

*Figure 1: Diversity Trajectories
(8 topology families, 30 seeds each)*

[PLACEHOLDER — original figure not available]

Fig. 1. Mean diversity trajectories (± 1 SE) for 8 topology families (30 seeds each). DAG families converge fastest; ring-skip2 ($\kappa = 47$) sustains diversity longest.

70% of diversity variance is explained by topology—a $14\times$ amplification over $K = 0$. The $K = 0$ control confirms these are genuine topology–landscape interactions.

High- κ topologies slow allele propagation, extending exploration. On OneMax this merely delays convergence; on rugged landscapes, sustained diversity prevents premature trapping. Ring-skip2 ($\kappa = 47$) maintains highest diversity but not highest fitness at $K = 6$ —bidir-ring ($\kappa = 10$) achieves marginally better fitness, suggesting a Goldilocks effect. HERA [17] independently converges to “compact chains with strategically preserved cycles”—intermediate κ under optimization pressure.

6 Discussion

At constant density, $\kappa(D)$ explains 17% of diversity variance and 24% of fitness variance at generation 30—medium-to-large effects on a landscape deliberately chosen to be easy. We interpret the temporal profile as *cycle-mediated exploration*: feedback loops slow the propagation of dominant alleles, extending the period during which subpopulations maintain independent search. On OneMax, selection pressure eventually overwhelms this phase. The NK pilot confirms: η^2 rises from 0.049 ($K = 0$) to 0.686 ($K = 6$), sustained above 0.50 at generation 500 while collapsing to near zero for $K = 0$.

Diversity as the primary channel. Across both experiments, diversity is most sensitive to topology. On NK landscapes, diversity $\eta^2 = 0.686$ while best-fitness

Figure 2: Pearson $r(\kappa, \text{diversity})$
over generations

[PLACEHOLDER — original figure not available]

Fig. 2. Pearson $r(\kappa, \text{diversity})$ over generations. Peak at gen. 20–50; decay by gen. 100.

$\eta^2 < 0.10$. Cycles operate through population-level diversity maintenance rather than direct fitness improvement. Practitioners should monitor diversity, not just fitness, when evaluating communication topologies.

The implications for MAS design are immediate. The systems in Table 1 concluded DAGs outperform, but by Theorem 1 their experiments compared sparse to dense. The correct conclusion is that *sparser communication is cheaper*, not that cycles are harmful. On the cost side, cycles do impose overhead: iterative message-passing until convergence, rather than the single-pass execution of DAGs. Our experiments quantify the *benefit* side—sustained diversity on rugged landscapes—completing a cost-benefit framework for topology choice.

Figure 6 illustrates the confound concretely. We ran the same island model GA on 13 cubic symmetric graphs from the Foster census [6] ($n = 4$ to 30, 30 seeds each). Because every 3-regular graph on n vertices has $m = 3n/2$ edges, the cycle rank $\beta_1 = n/2 + 1$, and the density $m/n = 3/2$ —all three quantities are affine in n , hence perfectly collinear. The left panel plots diversity against n (increasing trend); the right plots the *same data* against density $m/\binom{n}{2}$ (decreasing trend). Both narratives are the same regression—the trap Theorem 1 predicts. Only constant-density experiments break the degeneracy. As a positive control, the Dodecahedron and Desargues graphs share $n = 20$, $\beta_1 = 11$, density $3/19$; Mann–Whitney finds no significant difference ($p = 0.615$), confirming that graphs matched on all confounded variables produce indistinguishable dynamics.

Limitations. First, $n = 8$ islands is small; scaling experiments on larger graph families are needed (the Foster sweep uses up to $n = 30$ but cannot escape the density confound within the 3-regular family). Second, the Pearson correlation is marginal ($p = 0.063$) on 8 topology-level points, though significant at generation 50 ($r = 0.838$, $p = 0.009$) and corroborated by the 240-run ANOVA

Figure 3: κ vs mean diversity at generation 30

[PLACEHOLDER — original figure not available]

Fig. 3. κ vs. mean diversity at gen. 30. $r = -0.681$ ($p = 0.063$, 6 df).

($p < 10^{-7}$). Third, the NK pilot uses only 3 of 8 topologies and 10 seeds; a full-factorial experiment is needed for generality.

7 Related Work

Learned and adaptive topologies. Puppeteer [9] trains an RL orchestrator that discovers “compact, cyclic reasoning structures” on hard tasks, outperforming DAG-constrained MacNet [21]. Cycles *emerge* under optimization pressure, but no topological metric (β_1 , κ) is computed and no ablation separates cycles from density. By Theorem 1, their evidence is exactly the confound our experiment resolves. HERA [17] is the first system to explicitly track cycle count during topology evolution, converging to “compact chains with strategically preserved cycles”—intermediate κ under optimization pressure. Agent Q-Mix [14] learns six communication actions via QMIX, including a `debate_check` action that creates directed 2-cycles; it is the most topology-aware RL-based system to date. Our theorem explains why intermediate cycle counts work.

Joint prompt–topology optimization. MASS [29] is the closest predecessor: it jointly optimizes prompts and topologies for multi-agent systems via a three-stage pipeline (block-level prompts, workflow topology, workflow-level prompts). However, MASS restricts its topology search to DAGs, and its central finding—that “prompts matter more than topology”—may be an artifact of this restriction. By Theorem 1, varying DAG structure at constant node count simultaneously varies density. With cycles excluded from the search space, the dominant topological variable (κ) is held at zero, artificially suppressing topology’s explanatory power. Our experiments show that when κ is allowed to vary at constant density, topology explains up to 69% of diversity variance on rugged landscapes.

Figure 4: eta-squared for diversity and fitness over generations

[PLACEHOLDER — original figure not available]

Fig. 4. η^2 for diversity and fitness over generations. Peak near gen. 30; decay by gen. 100.

Large-scale studies. Lynch et al. [20] show that graph-parameterised clock systems constrain representable behaviours to paths in G : DAGs permit only acyclic execution. Dochkina et al. [10] conduct the largest controlled topology study to date: 25,000 tasks, 256 agents, and 8 coordination protocols yield a 44% quality spread (Cohen’s $d = 1.86$), demonstrating that topology choice is the dominant performance lever at industrial scale. Crucially, the effect is capability-moderated: weaker models benefit most from DAG structure, while stronger models exploit cyclic communication patterns—suggesting a Goldilocks interaction between agent capability and cycle count. However, sparsity and acyclicity are confounded in all runs; by Theorem 1, this is a density comparison. Our experiment provides the controlled condition their design lacks. Wang et al. [26] find that K -regular graphs maximise coordination quality, with a minimum capability threshold $\beta_{\min} \sim 1/K$: higher connectivity substitutes for individual agent intelligence. This highlights a coordination–exploration duality—density aids coordination but, as our experiments show, suppresses diversity-mediated exploration. The intersection defines an optimal topology that balances both.

Task structure and topology generation. Amir et al. [3] prove that Schur-convexity of a task’s performance function determines *whether* agent diversity improves outcomes, independent of the communication topology. Their result is the task-side dual of our framework: κ (and more broadly β_1) governs how much diversity the topology creates, while their Schur-convexity parameter σ governs whether that diversity is beneficial at all. Together, the two axes yield a practical design rule: measure σ first to decide if diversity helps, then tune κ to supply the right amount. Complementarily, TopoDIM [4] learns a task-to-topology mapping that generates communication graphs from task requirements, achieving 46% token savings over static topologies. This is the inverse of our analysis: we characterise how topology affects performance; they generate topology from performance

Figure 6: eta-squared by metric
across NK ruggedness (K)

[PLACEHOLDER — original figure not available]

Fig. 5. η^2 by metric across NK ruggedness (K) at gen. 200. Topology explains $< 5\%$ at $K = 0$ but $\sim 70\%$ of diversity at $K = 6$. Dashed: Cohen’s thresholds. $**p < 0.01$; $***p < 0.001$.

objectives. Our theoretical framework—particularly the density–cycle confound and the role of κ —could inform the topological invariants their learned functor targets.

Cycle structure. CycRak [22] shows short-cycle centrality outperforms eigenvector centrality for diffusion, suggesting cycle *length distribution* governs propagation—consistent with our non-monotonic bidir-ring result.

8 Conclusion

For connected graphs with fixed vertex count, cycle rank β_1 and edge count m differ by a constant (Theorem 1). Every prior comparison of cyclic versus acyclic topologies is therefore confounded. We escape this by moving to directed graphs, where $\kappa(D)$ is not determined by (n, m) . Across 240 runs at constant density, κ predicts diversity ($\eta^2 = 0.17$, $p < 10^{-7}$) and fitness ($\eta^2 = 0.24$). The effect is transient on OneMax but a $14\times$ amplification at NK $K = 6$ confirms scaling with landscape difficulty. Future work should pursue full-factorial NK experiments, κ –capability interaction, and κ as a practical topology design heuristic—precisely the directions called for by recent position papers [27].

References

1. Alba, E., Tomassini, M.: Parallelism and evolutionary algorithms. IEEE Transactions on Evolutionary Computation **6**(5), 443–462 (2002). <https://doi.org/10.1109/TEVC.2002.800880>

*Figure 5: Same data, opposite narrative
(Foster census cubic symmetric graphs)*

[PLACEHOLDER — original figure not available]

Fig. 6. Same data, opposite narrative. Left: diversity vs. n . Right: diversity vs. density $m/\binom{n}{2}$. 13 Foster census cubic symmetric graphs ($n = 4\text{--}30$, 30 seeds). For 3-regular graphs, β_1 , m , n are collinear—causal attribution is underdetermined.

2. Alba, E., Troya, J.M.: A survey of parallel distributed genetic algorithms. *Complexity* **4**(4), 31–52 (1999)
3. Amir, O.A., Bettini, M., Prorok, A.: Schur-convexity of task performance in multi-agent systems. *arXiv preprint arXiv:2506.09434* (2025)
4. Anonymous: TopoDIM: Topology-aware design of interactive multi-agent systems. *arXiv preprint arXiv:2601.10120* (2025)
5. Bei, Y., Zhang, W., Wang, S., Chen, W., Zhou, S., Chen, H., Li, Y., Bu, J., Pan, S., Yu, Y., King, I., Karray, F., Yu, P.S.: Graphs meet AI agents: Taxonomy, progress, and future opportunities. *arXiv preprint arXiv:2506.18019* (2025)
6. Bouwer, I.Z., Chernoff, W.W., Monson, B., Star, Z.: The Foster Census: R. M. Foster’s Census of Connected Symmetric Trivalent Graphs. Charles Babbage Research Centre (1988)
7. Cang, Y., Zhang, X., Zhao, E., Ji, Z., Liu, Y., He, Y., Ning, Z., Chen, Y., Que, W., Shi, L.: Graph-GRPO: Stabilizing multi-agent topology learning via group relative policy optimization. *arXiv preprint arXiv:2603.02701* (2026)
8. Chen, H., Zheng, X., Liu, Y., Jiao, P., Li, S., Liu, H., Zhao, Z., Xu, Z., Khalil, I., Pan, S.: GoAgent: Group-of-agents communication topology generation for LLM-based multi-agent systems. *arXiv preprint arXiv:2603.19677* (2026)
9. Dang, Y., Qian, C., Luo, X., Fan, J., Xie, Z., Shi, R., Chen, W., Yang, C., Che, X., Tian, Y., Xiong, X., Han, L., Liu, Z., Sun, M.: Multi-agent collaboration via evolving orchestration. In: *Advances in Neural Information Processing Systems* (NeurIPS) (2025)
10. Dochkina, V.: Drop the hierarchy and roles: How self-organizing LLM agents outperform designed structures. *arXiv preprint arXiv:2603.28990* (2026)
11. Hanks, T., Riess, H.J.: Cellular sheaves on graphs: A topological approach to data analysis. *arXiv preprint arXiv:2504.02049* (2025)

12. Hao, G., Dai, Y., Qin, X., Yu, S.: Brain-inspired graph multi-agent systems for LLM reasoning. arXiv preprint arXiv:2603.15371 (2026)
13. Hatcher, A.: Algebraic Topology. Cambridge University Press (2002)
14. Jiang, E.H., Li, L., Sun, R., Liang, X., Li, Y., Wu, Y., Luo, H., Li, H., Zhang, Z., Kang, Z., Chang, K.W., Wu, Y.N.: Agent Q-Mix: Selecting the right action for LLM multi-agent systems through reinforcement learning. arXiv preprint arXiv:2604.00344 (2026)
15. Johnson, D.B.: Finding all the elementary circuits of a directed graph. SIAM Journal on Computing **4**(1), 77–84 (1975). <https://doi.org/10.1137/0204007>
16. Kauffman, S.A.: The Origins of Order: Self-Organization and Selection in Evolution. Oxford University Press (1993)
17. Li, S., Ramakrishnan, N.: Experience as a compass: Multi-agent RAG with evolving orchestration and agent prompts. arXiv preprint arXiv:2604.00901 (2026)
18. Li, S., Liu, Y., Wen, Q., Zhang, C., Pan, S.: Assemble your crew: Automatic multi-agent communication topology design via autoregressive graph generation. In: Proceedings of the AAAI Conference on Artificial Intelligence (AAAI) (2026), oral presentation
19. Li, S., Liu, Y., Zheng, Y., Li, M., Nguyen, Q.V.H., Pan, S.: OFA-MAS: One-for-all multi-agent system topology design based on mixture-of-experts graph generative models. In: Proceedings of the ACM Web Conference (WWW). Dubai, United Arab Emirates (2026)
20. Lynch, O., Myers, D.J., Rischel, E.F., Staton, S.: Clock systems for stochastic and non-deterministic categorical systems theories. arXiv preprint arXiv:2603.29573 (2026)
21. Qian, C., Zhuge, M., Luo, X., Yang, C., Liu, Z., Sun, M.: MacNet: Multi-agent communication via orchestration within directed acyclic graphs. In: Proceedings of the International Conference on Machine Learning (2025)
22. Shi, W., Fan, T., Xu, S., Yang, R., Lü, L.: A universal optimization framework based on cycle ranking for influence maximization in complex networks. arXiv preprint arXiv:2405.09357 (2024)
23. Skolicki, Z., De Jong, K.: The influence of migration sizes and intervals on island models. In: Proceedings of the 2005 Genetic and Evolutionary Computation Conference (GECCO). pp. 1295–1302. ACM, Washington, DC, USA (2005). <https://doi.org/10.1145/1068009.1068219>
24. Tomassini, M.: Spatially Structured Evolutionary Algorithms. Springer (2005)
25. Wang, S., Lu, R., Yang, Z., Wang, Y., Zhang, Y., Xu, L., Xu, Q., Yin, G., Chen, C., Guan, X.: AgentConductor: Topology evolution for multi-agent competition-level code generation. arXiv preprint arXiv:2602.17100 (2026)
26. Wang, Z., Chen, H., Yang, J., Zhang, M., Wen, Q., Wang, J.: On the role of communication topology in multi-agent LLM systems: K-regular graphs maximize coordination. arXiv preprint arXiv:2604.07751 (2026)
27. Yang, J., Zhang, M., Jin, Y., Chen, H., Wen, Q., Lin, L., He, Y., Kumar, S., Xu, W., Evans, J., Wang, J.: Topological structure learning should be a research priority for LLM-based multi-agent systems. arXiv preprint arXiv:2505.22467 (2025)
28. Zhang, G., Yue, Y., Sun, X., Wan, G., Yu, M., Fang, J., Wang, K., Chen, T., Cheng, D.: G-Designer: Architecting multi-agent communication topologies via graph neural networks. In: Proceedings of the 42nd International Conference on Machine Learning (ICML) (2025)
29. Zhou, H., Wan, X., Sun, R., Palangi, H., Iqbal, S., Vulić, I., Korhonen, A., Arik, S.Ö.: Multi-agent design: Optimizing agents with better prompts and topologies. In:

Proceedings of the International Conference on Learning Representations (ICLR)
(2026)